



Cohort canonical evaluation — 105 agents

idea-01-lateral-attribution · 534 held-out segments · V0
yaw 0.01627 rad/s, CTE 254.26 m · generated
2026-06-03 00:38

- Executive summary

AGENTS (OK / TOTAL)

93 / 105

EVAL POOL

534 segs

V0 YAW RMSE

0.01627

V0 CTE RMSE

254.26 m

WALL TIME

40.33s

Best yaw: module-4.v2_agent-04 **+57.9%** · Best CTE:

module-3.v3_agent-03 **+72.3%**

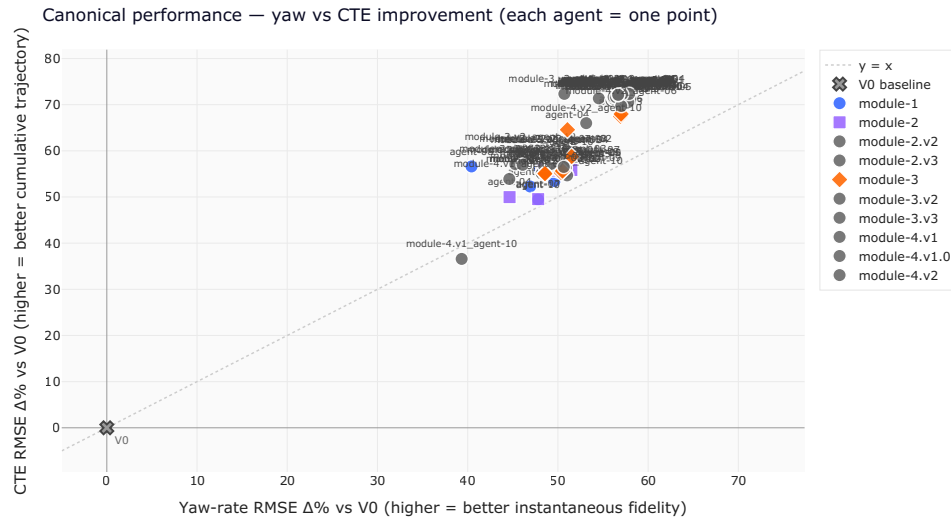
Winning both KPIs $\geq$ +30% (93 agents):

module-4.v2_agent-04, module-4.v1.01_agent-05,
module-4.v1_agent-06, module-4.v1_agent-01,
module-4.v1.01_agent-04, module-3.v3_agent-06,
module-4.v1_agent-04, module-4.v2_agent-05,
module-4.v2_agent-06, module-3.v3_agent-04, m3-agent-06,
module-4.v1.01_agent-01, m3-agent-03,
module-3.v2_agent-04, module-3.v3_agent-08,
module-3.v2_agent-09, module-3.v3_agent-10,
module-3.v2_agent-02, module-3.v2_agent-03,
module-4.v2_agent-08, module-4.v2_agent-07, m3-agent-01,
module-3.v2_agent-05, module-3.v2_agent-06,
module-3.v2_agent-07, module-3.v2_agent-08,
module-3.v2_agent-10, module-4.v2_agent-01,
module-4.v2_agent-03, module-3.v3_agent-07,
module-3.v3_agent-09, module-4.v1_agent-02,
module-4.v1_agent-09, module-4.v1_agent-03,
module-3.v3_agent-02, module-4.v1.01_agent-03,
module-4.v2_agent-02, module-4.v1_agent-08,
module-4.v1_agent-05, module-3.v3_agent-01,
module-3.v3_agent-05, module-3.v2_agent-01,
module-4.v2_agent-10, m3-agent-10, m2-agent-05, m3-agent-04,
module-2.v3_agent-10, m1-agent-01,
module-2.v2_agent-09, module-2.v3_agent-09,
module-2.v3_agent-06, module-3.v3_agent-03,
module-2.v3_agent-07, module-4.v2_agent-09, m3-agent-09,
m3-agent-08, m2-agent-08, m1-agent-09,
module-2.v3_agent-02, module-2.v2_agent-05, m1-agent-07,
module-2.v3_agent-04, module-2.v2_agent-08,
module-2.v3_agent-03, m2-agent-10, module-2.v2_agent-04,

module-2.v3_agent-05, m1-agent-02, m1-agent-05, m2-agent-01, module-2.v2_agent-06, m3-agent-07, m3-agent-02, module-2.v2_agent-02, m2-agent-03, m1-agent-03, m2-agent-02, module-2.v3_agent-01, m2-agent-09, m2-agent-06, module-2.v2_agent-01, m1-agent-04, m2-agent-07, m1-agent-10, module-2.v2_agent-07, m1-agent-06, module-2.v3_agent-08, module-2.v2_agent-10, module-2.v2_agent-03, m2-agent-04, module-4.v1_agent-07, m1-agent-08, module-4.v1_agent-10

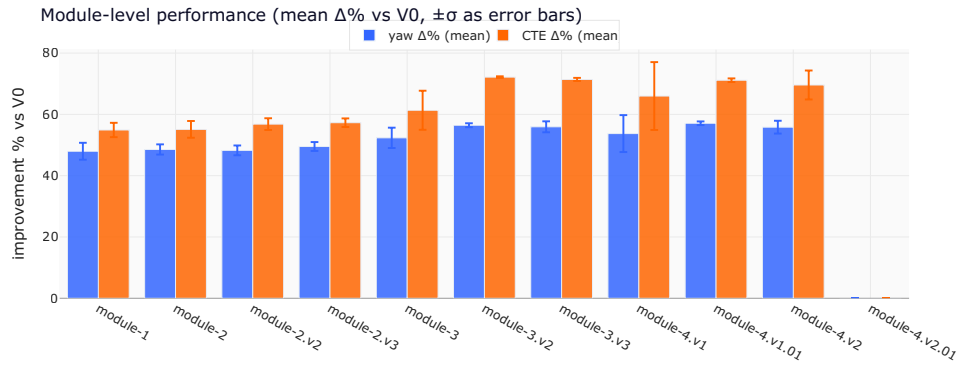
1. Headline scatter — yaw vs CTE improvement

Each agent is one point. **Top-right is the goal** — both KPIs improve over V0. The dashed diagonal is $y = x$: above it, an agent improved CTE more than yaw (good trajectory integration); below, the opposite. The V0 baseline sits at the origin.



2. Module-level aggregation

Performance grouped by which module shipped the agent.
Mean $\Delta\%$ over only the agents that reconstructed successfully;
failures shown separately.



family	n ok / total	yaw $\Delta\%$ (mean $\pm \sigma$)	CTE $\Delta\%$ (mean $\pm \sigma$)
module-1	10/10	+48.0% $\pm$ 2.8%	+55.0% $\pm$ 2.0%
module-2	10/10	+48.6% $\pm$ 1.7%	+55.0% $\pm$ 1.5%
module-2.v2	10/10	+48.3% $\pm$ 1.6%	+57.0% $\pm$ 1.5%
module-2.v3	10/10	+49.5% $\pm$ 1.4%	+57.0% $\pm$ 1.5%
module-3	9/10	+52.3% $\pm$ 3.3%	+61.0% $\pm$ 2.5%
module-3.v2	10/10	+56.5% $\pm$ 0.6%	+72.0% $\pm$ 1.0%
module-3.v3	10/10	+55.9% $\pm$ 1.8%	+71.0% $\pm$ 1.5%
module-4.v1	10/10	+53.7% $\pm$ 6.0%	+66.0% $\pm$ 4.0%
module-4.v1.01	4/5	+57.1% $\pm$ 0.6%	+71.0% $\pm$ 1.5%
module-4.v2	10/10	+55.8% $\pm$ 2.1%	+69.0% $\pm$ 2.5%
module-4.v2.01	0/10	—	—

- **2b. Token expenditure per agent**

Sourced from each agent's Claude Code subagent transcript (~/.claude/projects/<proj>/*/subagents/agent-*.jsonl). Tokens summed across every assistant turn of the latest-mtime run per agent.

COHORT TOTAL

331.52M

MEDIAN / AGENT

3.16M

MEDIAN TURNS

64

AGENTS W/ TRANSCRIPT

95

By family

family	n	total tokens	median / agent	me
module-1	10	23.38M	2.38M	
module-2	10	25.69M	2.24M	

family	n	total tokens	median / agent	me
module-2.v2	10	30.22M	3.13M	
module-2.v3	10	37.55M	3.92M	
module-3	10	41.38M	3.50M	
module-3.v2	10	43.93M	4.49M	
module-3.v3	10	57.27M	5.50M	
module-4.v1	10	28.54M	2.43M	
module-4.v1.01	5	11.43M	2.18M	
module-4.v2	10	32.11M	3.14M	
module-4.v2.01	0	—	—	

Per agent (sorted by total)

agent	family	turns	total
module-3.v3_agent-07	module-3.v3	109	9.24
m3-agent-06	module-3	107	8.18
module-3.v3_agent-03	module-3.v3	113	7.36
module-3.v3_agent-06	module-3.v3	98	7.29
m3-agent-01	module-3	94	6.78
module-3.v3_agent-09	module-3.v3	97	6.42
module-3.v3_agent-01	module-3.v3	84	6.22

agent	family	turns	total
m3-agent-05	module-3	106	6.03
module-3.v2_agent-03	module-3.v2	81	5.52
module-4.v1_agent-09	module-4.v1	97	5.29
module-3.v2_agent-06	module-3.v2	78	5.16
module-3.v2_agent-02	module-3.v2	76	4.89
module-4.v2_agent-03	module-4.v2	80	4.85
module-3.v2_agent-01	module-3.v2	70	4.83
module-3.v3_agent-10	module-3.v3	70	4.79
module-2.v3_agent-05	module-2.v3	73	4.79
module-3.v2_agent-05	module-3.v2	75	4.75
module-3.v3_agent-04	module-3.v3	75	4.47
m2-agent-05	module-2	83	4.45
module-2.v3_agent-01	module-2.v3	67	4.43
module-4.v1_agent-04	module-4.v1	85	4.36
module-2.v3_agent-03	module-2.v3	67	4.26
module-3.v2_agent-04	module-3.v2	64	4.23
module-3.v2_agent-09	module-3.v2	69	4.20
module-3.v3_agent-08	module-3.v3	70	4.13
module-3.v3_agent-05	module-3.v3	60	4.06

agent	family	turns	total
module-3.v2_agent-10	module-3.v2	66	4.03
module-2.v3_agent-06	module-2.v3	73	4.01
module-2.v3_agent-02	module-2.v3	69	3.98
m3-agent-03	module-3	72	3.96
module-2.v3_agent-04	module-2.v3	67	3.86
module-4.v2_agent-08	module-4.v2	81	3.85
m3-agent-08	module-3	68	3.84
m2-agent-08	module-2	76	3.73
module-3.v2_agent-07	module-3.v2	58	3.68
module-2.v2_agent-07	module-2.v2	56	3.63
module-2.v3_agent-07	module-2.v3	69	3.61
m2-agent-07	module-2	69	3.61
module-4.v2_agent-05	module-4.v2	62	3.56
m1-agent-09	module-1	79	3.54
module-2.v2_agent-01	module-2.v2	64	3.53
module-2.v3_agent-10	module-2.v3	66	3.41
module-3.v3_agent-02	module-3.v3	70	3.29
module-2.v2_agent-04	module-2.v2	61	3.29
module-4.v2_agent-01	module-4.v2	69	3.27

agent	family	turns	total
module-2.v2_agent-08	module-2.v2	59	3.19
module-4.v2_agent-09	module-4.v2	72	3.18
m3-agent-07	module-3	66	3.16
module-2.v2_agent-03	module-2.v2	58	3.14
module-4.v2_agent-07	module-4.v2	63	3.11
module-2.v2_agent-02	module-2.v2	54	3.11
module-4.v1_agent-08	module-4.v1	68	3.08
module-4.v2_agent-04	module-4.v2	66	3.03
m3-agent-09	module-3	61	3.02
module-2.v2_agent-09	module-2.v2	57	3.00
module-4.v2_agent-02	module-4.v2	58	3.00
m1-agent-08	module-1	74	2.96
module-4.v1_agent-05	module-4.v1	65	2.90
module-2.v2_agent-05	module-2.v2	54	2.88
m3-agent-10	module-3	65	2.86
m3-agent-02	module-3	61	2.83
module-4.v1.01_agent-03	module-4.v1.01	60	2.81
module-4.v1.01_agent-05	module-4.v1.01	61	2.78
module-2.v3_agent-09	module-2.v3	56	2.72

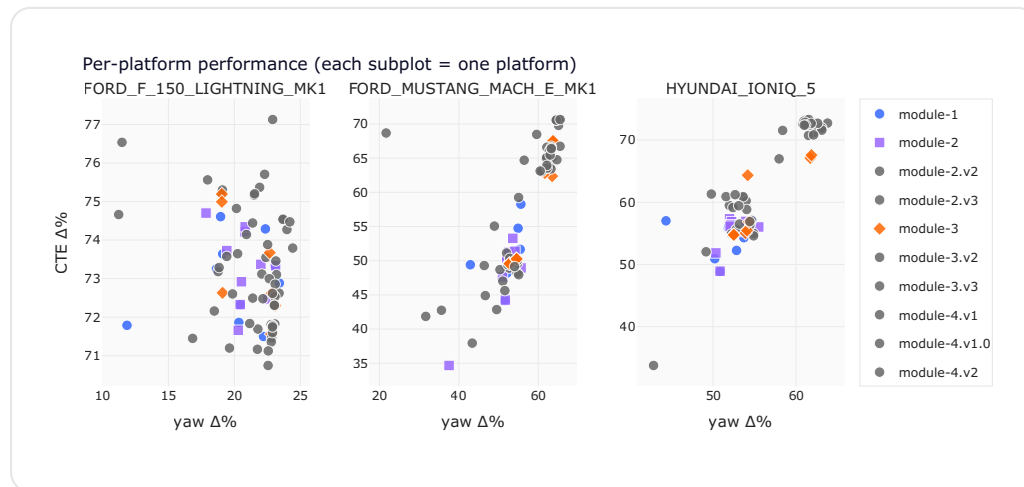
agent	family	turns	total
m2-agent-02	module-2	68	2.68
m1-agent-02	module-1	66	2.66
module-3.v2_agent-08	module-3.v2	47	2.63
module-4.v1_agent-02	module-4.v1	54	2.50
module-2.v3_agent-08	module-2.v3	55	2.49
m1-agent-10	module-1	64	2.45
module-4.v2_agent-10	module-4.v2	58	2.44
m1-agent-07	module-1	62	2.41
module-2.v2_agent-06	module-2.v2	48	2.37
module-4.v1_agent-03	module-4.v1	49	2.36
m2-agent-03	module-2	52	2.35
m1-agent-04	module-1	63	2.34
module-4.v1_agent-10	module-4.v1	44	2.29
m1-agent-05	module-1	62	2.25
module-4.v1_agent-01	module-4.v1	46	2.18
module-4.v1.01_agent-04	module-4.v1.01	51	2.18
m2-agent-09	module-2	55	2.14
m2-agent-10	module-2	53	2.11
module-2.v2_agent-10	module-2.v2	42	2.07

agent	family	turns	total
module-4.v1_agent-06	module-4.v1	51	2.04
module-4.v1.01_agent-01	module-4.v1.01	43	1.93
m1-agent-01	module-1	53	1.88
module-4.v2_agent-06	module-4.v2	44	1.83
m2-agent-04	module-2	45	1.83
module-4.v1.01_agent-02	module-4.v1.01	44	1.73
m1-agent-06	module-1	54	1.73
m2-agent-06	module-2	43	1.66
module-4.v1_agent-07	module-4.v1	39	1.55
m1-agent-03	module-1	38	1.17
m2-agent-01	module-2	41	1.13
m3-agent-04	module-3	26	740.2

• 3. Per-platform breakdown

Same cohort, but split by the vehicle platform the agent was evaluated on. Agents that win one platform but lose another

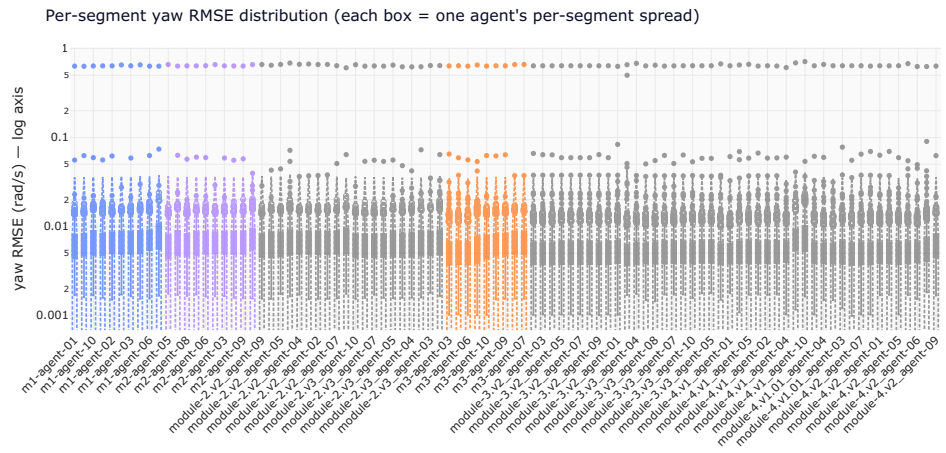
have a platform-specific calibration; agents whose dots cluster together generalise.



platform	agents	yaw Δ% (mean ± σ)
FORD_F_150_LIGHTNING_MK1	93	+21.4% ± 2.5%
FORD_MUSTANG_MACH_E_MK1	93	+56.6% ± 8.0%
HYUNDAI_IONIQ_5	93	+56.7% ± 4.8%

• 4. Per-segment yaw RMSE distribution

Each box summarises one agent's per-segment yaw RMSE — pooled RMSE can hide that an agent does great on most segments but pathologically badly on a few. The y-axis is logarithmic to compress outliers.



5. Per-agent canonical scorecard

The full table. Failed reconstructions are surfaced with their reason — no fake zeros.

agent	family	status
m1-agent-01	module-1	ok
m1-agent-02	module-1	ok
m1-agent-03	module-1	ok
m1-agent-04	module-1	ok
m1-agent-05	module-1	ok
m1-agent-06	module-1	ok
m1-agent-07	module-1	ok
m1-agent-08	module-1	ok
m1-agent-09	module-1	ok
m2-agent-01	module-2	ok
m2-agent-02	module-2	ok
m2-agent-03	module-2	ok
m2-agent-04	module-2	ok
m2-agent-05	module-2	ok
m2-agent-06	module-2	ok
m2-agent-07	module-2	ok
m2-agent-08	module-2	ok
m2-agent-09	module-2	ok
m3-agent-01	module-3	ok
m3-agent-02	module-3	ok
m3-agent-03	module-3	ok
m3-agent-04	module-3	ok
m3-agent-05	module-3	ok
m3-agent-06	module-3	ok
m3-agent-07	module-3	ok
m3-agent-08	module-3	ok
m3-agent-09	module-3	ok
module-1-v1	module-1	ok
module-1-v2	module-1	ok
module-2-v1	module-2	ok
module-2-v2	module-2	ok
module-3-v1	module-3	ok
module-3-v2	module-3	ok
module-4-v1	module-4	ok
module-4-v2	module-4	ok

agent	family	status
m1-agent-06	module-1	ok
m1-agent-07	module-1	ok
m1-agent-08	module-1	ok
m1-agent-09	module-1	ok
m1-agent-10	module-1	ok
m2-agent-01	module-2	ok
m2-agent-02	module-2	ok
m2-agent-03	module-2	ok
m2-agent-04	module-2	ok
m2-agent-05	module-2	ok
m2-agent-06	module-2	ok
m2-agent-07	module-2	ok
m2-agent-08	module-2	ok
m2-agent-09	module-2	ok
m2-agent-10	module-2	ok
module-2.v2_agent-01	module-2.v2	ok
module-2.v2_agent-02	module-2.v2	ok

agent	family	status
module-2.v2_agent-03	module-2.v2	ok
module-2.v2_agent-04	module-2.v2	ok
module-2.v2_agent-05	module-2.v2	ok
module-2.v2_agent-06	module-2.v2	ok
module-2.v2_agent-07	module-2.v2	ok
module-2.v2_agent-08	module-2.v2	ok
module-2.v2_agent-09	module-2.v2	ok
module-2.v2_agent-10	module-2.v2	ok
module-2.v3_agent-01	module-2.v3	ok
module-2.v3_agent-02	module-2.v3	ok
module-2.v3_agent-03	module-2.v3	ok
module-2.v3_agent-04	module-2.v3	ok
module-2.v3_agent-05	module-2.v3	ok
module-2.v3_agent-06	module-2.v3	ok
module-2.v3_agent-07	module-2.v3	ok
module-2.v3_agent-08	module-2.v3	ok
module-2.v3_agent-09	module-2.v3	ok

agent	family	status
module-2.v3_agent-10	module-2.v3	ok
m3-agent-01	module-3	ok
m3-agent-02	module-3	ok
m3-agent-03	module-3	ok
m3-agent-04	module-3	ok
m3-agent-05	module-3	import_
m3-agent-06	module-3	ok
m3-agent-07	module-3	ok
m3-agent-08	module-3	ok
m3-agent-09	module-3	ok
m3-agent-10	module-3	ok
module-3.v2_agent-01	module-3.v2	ok
module-3.v2_agent-02	module-3.v2	ok
module-3.v2_agent-03	module-3.v2	ok
module-3.v2_agent-04	module-3.v2	ok
module-3.v2_agent-05	module-3.v2	ok
module-3.v2_agent-06	module-3.v2	ok

agent	family	status
module-3.v2_agent-07	module-3.v2	ok
module-3.v2_agent-08	module-3.v2	ok
module-3.v2_agent-09	module-3.v2	ok
module-3.v2_agent-10	module-3.v2	ok
module-3.v3_agent-01	module-3.v3	ok
module-3.v3_agent-02	module-3.v3	ok
module-3.v3_agent-03	module-3.v3	ok
module-3.v3_agent-04	module-3.v3	ok
module-3.v3_agent-05	module-3.v3	ok
module-3.v3_agent-06	module-3.v3	ok
module-3.v3_agent-07	module-3.v3	ok
module-3.v3_agent-08	module-3.v3	ok
module-3.v3_agent-09	module-3.v3	ok
module-3.v3_agent-10	module-3.v3	ok
module-4.v1_agent-01	module-4.v1	ok
module-4.v1_agent-02	module-4.v1	ok
module-4.v1_agent-03	module-4.v1	ok

agent	family	status
module-4.v1_agent-04	module-4.v1	ok
module-4.v1_agent-05	module-4.v1	ok
module-4.v1_agent-06	module-4.v1	ok
module-4.v1_agent-07	module-4.v1	ok
module-4.v1_agent-08	module-4.v1	ok
module-4.v1_agent-09	module-4.v1	ok
module-4.v1_agent-10	module-4.v1	ok
module-4.v1.01_agent-01	module-4.v1.01	ok
module-4.v1.01_agent-02	module-4.v1.01	missing_manifest
module-4.v1.01_agent-03	module-4.v1.01	ok
module-4.v1.01_agent-04	module-4.v1.01	ok
module-4.v1.01_agent-05	module-4.v1.01	ok
module-4.v2_agent-01	module-4.v2	ok
module-4.v2_agent-02	module-4.v2	ok
module-4.v2_agent-03	module-4.v2	ok
module-4.v2_agent-04	module-4.v2	ok
module-4.v2_agent-05	module-4.v2	ok

agent	family	status
module-4.v2_agent-06	module-4.v2	ok
module-4.v2_agent-07	module-4.v2	ok
module-4.v2_agent-08	module-4.v2	ok
module-4.v2_agent-09	module-4.v2	ok
module-4.v2_agent-10	module-4.v2	ok
module-4.v2.01_agent-01	module-4.v2.01	missing_manifest
module-4.v2.01_agent-02	module-4.v2.01	missing_manifest
module-4.v2.01_agent-03	module-4.v2.01	missing_manifest
module-4.v2.01_agent-04	module-4.v2.01	missing_manifest
module-4.v2.01_agent-05	module-4.v2.01	missing_manifest
module-4.v2.01_agent-06	module-4.v2.01	missing_manifest
module-4.v2.01_agent-07	module-4.v2.01	missing_manifest
module-4.v2.01_agent-08	module-4.v2.01	missing_manifest
module-4.v2.01_agent-09	module-4.v2.01	missing_manifest
module-4.v2.01_agent-10	module-4.v2.01	missing_manifest

- **6. Calibration cards — fitted coefficients**

Where the cohort converges on similar physics, and where it forks. Each tile is one agent's `coeffs.json` as shipped.

m1-agent-02

```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.4760851016953556,
    "K_us": 0.0026722138032651867,
    "bias_rad": 0.0002686192499419401
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.809760457416018,
    "K_us": 0.0038412171248315657,
    "bias_rad": 0.001307379955568698
  },
  "HYUNDAI_IONIQ_5": {
    "L": 3.096229822470566,
    "K_us": 0.003695564361835472,
    "bias_rad": -0.0002960976826300522
  },
  "TESLA_MODEL_3": {
    "L": 2.875,
    "K_us": 0.0,
    "bias_rad": 0.0
  }
}
```

m1-agent-04

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "a": -0.004329037631466362,
    "b": 0.9177897626357783,
    "c": -0.00044178131403551616
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "a": 0.0002206757945255914,
    "b": 1.1513559927303039,
    "c": -0.0005583363495612761
  },
  "HYUNDAI_IONIQ_5": {
    "a": 0.0020222989265629857,
    "b": 0.9120045284217976,
    "c": -0.00048449411444108955
  },
  "TESLA_MODEL_3": {
    "a": 0.0002206757945255914,
    "b": 1.1513559927303039,
    "c": -0.0005583363495612761
  },
  "_default": {
    "a": 0.0,
    "b": 1.0,
    "c": 0.0
  }
}
```

m1-agent-06

```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.984,
    "best": "M2_understeer",
    "m1": {
      "a": 1.0870612234849137
    },
    "m2": {
      "L_eff": 2.4763030468069056,
      "K_us": 0.0024395167795073017
    },
    "m3": {
      "a": 0.9751496691433119,
      "b": 0.23839497273290036,
      "c": 0.0004082870644824788
    },
    "val_rmse": {
      "V0": 0.0267999381028469,
      "M1": 0.027439581036525294,
      "M2": 0.026420879339395156,
      "M3": 0.031903168689021696
    },
    "train_rmse": {
      "V0": 0.012685628695945105,
      "M1": 0.011389013310887795,
      "M2": 0.008745921115146504,
      "M3": 0.010386863485176999
    }
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "best": "M2_understeer",
    "m1": {
      "a": 0.8569324797501493
    },
  },
}
```

```
"m2": {
  "L_eff": 3.882198880475702,
  "K_us": 0.003439279945738158
},
"m3": {
  "a": 0.8772240685582882,
  "b": -0.031353830652592465,
  "c": -0.0036354381262862774
},
"val_rmse": {
  "V0": 0.01591104278075153,
  "M1": 0.011141896721881754,
  "M2": 0.0073696198273352906,
  "M3": 0.0104650790814036
},
"train_rmse": {
  "V0": 0.02018452928972004,
```

m1-agent-08

```
{
  "HYUNDAI_IONIQ_5": {
    "L": 3.0,
    "K": 0.0033895612010179177,
    "s": 0.9668278221152291,
    "off": 0.0005045046778856075,
    "lag": 3,
    "rmse": 0.00888482422851739
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.984,
    "K": 0.002625595512351926,
    "s": 1.1771921499482927,
    "off": 3.9923690213217876e-05,
    "lag": 4,
    "rmse": 0.01315660341959157
  },
}
```



```
"FORD_F_150_LIGHTNING_MK1": {
  "L": 3.7,
  "K": 0.0035388455508032707,
  "s": 0.9569361755429138,
  "off": -0.0011908292677060655,
  "lag": 3,
  "rmse": 0.011832340064259452
},
"TESLA_MODEL_3": {
  "L": 2.875,
  "K": 0.0033895612010179177,
  "s": 1.0,
  "off": 0.0,
  "lag": 3,
  "rmse": null
}
}
```

m1-agent-10

```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "L_eff": 2.5297190644305063,
    "K_us": 0.0015446536838391765,
    "d0": -4.756112262525493e-05,
    "tau_yaw": 0.02,
    "tau_steer": 0.05
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L_eff": 3.834827559665228,
    "K_us": 0.0031524889056087045,
    "d0": 0.00115199881391885,
```

```
    "tau_yaw": 0.0,
    "tau_steer": 0.05
  },
  "HYUNDAI_IONIQ_5": {
    "L_eff": 3.1485752690608293,
    "K_us": 0.0021732037046845788,
    "d0": -0.0005619614330402095,
    "tau_yaw": 0.0,
    "tau_steer": 0.05
  },
  "TESLA_MODEL_3": {
    "_note": "no truth available \u2014 passthrough yaw_rate_pred.",
    "passthrough": true
  }
}
```

m2-agent-03

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "platform": "FORD_F_150_LIGHTNING_MK1",
    "L": 3.7,
```

```
"n_samples": 430990,  
"rmse_v0": 0.016327157090019005,  
"rmse_v1": 0.007867193883977044,  
"K_v1": 0.0043769531249999985,  
"rmse_v2": 0.00640773025816659,  
"K_v2": 0.003836687017912508,  
"a_v2": 0.9743406550510199,  
"b_v2": -0.001228592917784131  
},  
"FORD_MUSTANG_MACH_E_MK1": {  
  "platform": "FORD_MUSTANG_MACH_E_MK1",  
  "L": 2.984,  
  "n_samples": 615175,  
  "rmse_v0": 0.013616820108542612,  
  "rmse_v1": 0.013930807470187448,  
  "K_v1": 0.000785400390624989,  
  "rmse_v2": 0.0095149036095479,  
  "K_v2": 0.0028742800060782272,  
  "a_v2": 1.1998919734409932,  
  "b_v2": 3.385609342398125e-05  
},  
"HYUNDAI_IONIQ_5": {  
  "platform": "HYUNDAI_IONIQ_5",  
  "L": 3.0,  
  "n_samples": 2042270,  
  "rmse_v0": 0.017084454875116246,  
  "rmse_v1": 0.009100616867655459,  
  "K_v1": 0.0042265624999999985,  
  "rmse_v2": 0.008741269113389604,  
  "K_v2": 0.0033859035164445724,  
  "a_v2": 0.9707125684779052,  
  "b_v2": 0.0005066770839987745  
}  
}
```

m2-agent-07

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "tau": 0.05,
    "L_eff": 3.8588781722695416,
    "K_u": 0.00314039574481963,
    "delta_bias_rad": -0.001157331673169034,
    "rmse": 0.005592721659455879
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "tau": 0.08,
    "L_eff": 2.5486304264900417,
    "K_u": 0.0015312706121448161,
    "delta_bias_rad": 4.8217638587095315e-05,
    "rmse": 0.008347407485517062
  },
  "HYUNDAI_IONIQ_5": {
    "tau": 0.05,
    "L_eff": 3.163750474852756,
    "K_u": 0.0021607848879418314,
    "delta_bias_rad": 0.0005617097326308059,
    "rmse": 0.007713638558960502
  }
}
```

m3-agent-01

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 0.863,
    "L_eff": 3.26,
    "K_us": 0.0035,
    "tau": 0.06,
    "delta0": 0.00133,
    "use_per_segment_delta0": false
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 0.891,
    "L_eff": 2.22,
    "K_us": 0.0015,
    "tau": 0.069,
    "delta0": -0.0001,
    "use_per_segment_delta0": true
  },
  "HYUNDAI_IONIQ_5": {
    "g": 0.93817,
    "L_eff": 2.8871,
    "K_us": 0.0028925,
    "tau": 0.061895,
    "delta0": 0.0,
    "use_per_segment_delta0": true
  },
  "_meta": {
    "delta0_detector": {
      "method": "yaw_rate_pred_rads_low_speed_high",
      "yr_thresh": 0.03,
      "v_thresh": 5.0,
      "min_rows": 50
    },
  },
}
```

```
    "notes": "Lightning coeffs hand-set from anti-patterns.md prior to 2020-08-01"
  }
}
```

m3-agent-03

```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 1.031539710821034,
    "L_eff": 2.605202152510245,
    "K_us": 0.001860730244989818,
    "tau": 0.0627605678289049,
    "delta0_fallback": 0.00016924506921824425,
    "v_thresh": 8.0,
    "delta_thresh": 0.005,
    "min_rows": 80,
    "use_per_segment_delta0": true
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 0.8860740528359514,
    "L_eff": 3.380345065967063,
    "K_us": 0.0032867081270640908,
    "tau": 0.05844688797895399,
    "delta0": 0.001336241196109372,
    "use_per_segment_delta0": false
  },
  "HYUNDAI_IONIQ_5": {
    "g": 0.896091628794913,
    "L_eff": 2.8008769715764292,
    "K_us": 0.0023750439062653513,
    "tau": 0.04533958799360384,
    "delta0_fallback": -0.00028615565156215293,
    "v_thresh": 8.0,
    "delta_thresh": 0.005,
    "min_rows": 80,
    "use_per_segment_delta0": true
  }
}
```

m3-agent-06

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "use_per_segment_delta0": false,
    "delta0": 0.0010739940929899243,
    "g": 0.8428393582368884,
    "L_eff": 3.26,
    "K_us": 0.0024711101464802783,
    "tau": 0.05493261284467375
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "use_per_segment_delta0": true,
    "delta0_fallback": -9.587089484890604e-05,
    "delta0": -9.587089484890604e-05,
    "g": 0.8814740811859291,
    "L_eff": 2.22,
    "K_us": 0.001600317906539393,
    "tau": 0.08022498681872611
  },
  "HYUNDAI_IONIQ_5": {
    "use_per_segment_delta0": true,
    "delta0_fallback": 7.704255037895423e-05,
    "delta0": 7.704255037895423e-05,
    "g": 0.9863062253350734,
    "L_eff": 3.033023238719065,
    "K_us": 0.0028341308076772504,
    "tau": 0.055317906584940124
  },
  "TESLA_MODEL_3": {
    "passthrough": true
  }
}
```

m3-agent-08

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 0.6006343437133439,
    "delta0": 0.0012709657890603524,
    "K_us": 0.002411110430682984,
    "tau": 0.05905789911423279,
    "L_eff": 2.273362244343053,
    "loss": 0.005658956187762647,
    "n_segments_fit": 175
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 0.9449514060826516,
    "delta0": -2.802364842758471e-05,
    "K_us": 0.0023030236477064245,
    "tau": 0.06905315476018789,
    "L_eff": 2.3418351435366787,
    "loss": 0.008957550273019014,
    "n_segments_fit": 240
  },
  "HYUNDAI_IONIQ_5": {
    "g": 0.8635558962089673,
    "delta0": -0.0002751563706215791,
    "K_us": 0.0031040456441715804,
    "tau": 0.051877028201724164,
    "L_eff": 2.663564560241335,
    "loss": 0.009441409401572077,
    "n_segments_fit": 400
  }
}
```


m3-agent-10

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 1.2677013760203653,
    "L_eff": 4.8869672279395875,
    "K_us": 0.004638743872488176,
    "tau": 0.056604854781974856,
    "delta0_glob": 0.0012307692372779105,
    "use_per_segment_delta0": false,
    "train_yaw_rmse": 0.012681866822813753
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 1.4999996985088524,
    "L_eff": 3.7976856683656175,
    "K_us": 0.0030911056729474493,
    "tau": 0.06768316418910694,
    "delta0_glob": 0.0003094591239113909,
    "use_per_segment_delta0": true,
    "train_yaw_rmse": 0.013524463171650382
  },
  "HYUNDAI_IONIQ_5": {
    "g": 1.4070879006714407,
    "L_eff": 4.3500385986664405,
    "K_us": 0.004915645105647427,
    "tau": 0.05040893663177004,
    "delta0_glob": -0.0005219437645709744,
    "use_per_segment_delta0": false,
    "train_yaw_rmse": 0.009412230675873393
  }
}
```

module-2.v2_agent-02

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "k": 0.9366323579288449,
    "K_us": 0.0008734759361408742,
    "b": -0.004441819259140514
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "k": 1.1760967499879564,
    "K_us": 0.0008734571660548825,
    "b": 0.0001733616708132899
  },
  "HYUNDAI_IONIQ_5": {
    "k": 0.9342137556967759,
    "K_us": 0.0009874770996559546,
    "b": 0.002002086108887664
  },
  "TESLA_MODEL_3": {
    "k": 1.0,
    "K_us": 0.0,
    "b": 0.0
  }
}
```

module-2.v2_agent-04

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "d0": 0.0012435085448964291,
    "kk": 0.9364948189665847,
    "KK": 0.0032470115980856545,
    "tau": 0.0
  },
  "HYUNDAI_IONIQ_5": {
    "L": 2.97,
    "d0": -0.0005236657821362199,
```

```
    "kk": 0.933574693892375,  
    "KK": 0.0029030763922389792,  
    "tau": 0.0  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "L": 2.984,  
    "d0": -2.675943355525215e-05,  
    "kk": 1.1761809849040472,  
    "KK": 0.0026070689183535123,  
    "tau": 0.0  
  },  
  "TESLA_MODEL_3": {  
    "L": 2.875,  
    "d0": 0.0,  
    "kk": 1.0,  
    "KK": 0.0,  
    "tau": 0.0  
  }  
}
```

module-2.v2_agent-06

```
{  
  "FORD_F_150_LIGHTNING_MK1": {  
    "a": 0.9397533271985806,  
    "K": 0.0008518149458872777,  
    "b": -0.0044345248756529525  
  },  
}
```

```
"FORD_MUSTANG_MACH_E_MK1": {
  "a": 1.1808972441892185,
  "K": 0.0008794604321610935,
  "b": 0.00016328778675285131
},
"HYUNDAI_IONIQ_5": {
  "a": 0.9350770580282316,
  "K": 0.0009883221917463027,
  "b": 0.002004778269326024
},
"TESLA_MODEL_3": {
  "a": 1.0,
  "K": 0.0,
  "b": 0.0
}
}
```

module-2.v2_agent-08

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "G": 0.9041237868107276,
    "Kus": 0.0007955703708621087,
    "bias": -0.005088051550305564,
    "tau": 0.05804217686893913,
    "kdot": -0.012302023150477267
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "G": 1.1560312029550013,
    "Kus": 0.0010408901778361697,
    "bias": 0.0007331723308643077,
    "tau": 0.040688651274880756,
    "kdot": 0.008800778388725197
  },
}
```

```
"HYUNDAI_IONIQ_5": {
  "G": 0.9700159639975359,
  "Kus": 0.001582087902935487,
  "bias": 0.00211745029940103,
  "tau": 0.04998898336470954,
  "kdot": 3.516097849258846e-05
},
"TESLA_MODEL_3": {
  "G": 1.0,
  "Kus": 0.0,
  "bias": 0.0,
  "tau": 0.0,
  "kdot": 0.0
}
}
```

module-2.v2_agent-10

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "k": 0.0009299897686168917,
    "g": -0.13146377934999154,
    "b": -0.004858365647164137
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "k": 0.0007776286604362939,
    "g": 0.1760123050495998,
    "b": 0.0007151284926824161
  },
  "HYUNDAI_IONIQ_5": {
    "k": 0.0017086888155224455,
    "g": -0.02937372993329363,
    "b": 0.0030989091566628654
  },
  "TESLA_MODEL_3": {
    "k": 0.0,
    "g": 0.0,
    "b": 0.0
  }
}
```

```
}  
}
```

module-2.v3_agent-02

```
{  
  "FORD_F_150_LIGHTNING_MK1": {  
    "L_eff": 3.941941906538264,  
    "K_us": 0.0037788233688761494,  
    "b": -0.005450661568006636,  
    "tau": -0.08207658150714123  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "L_eff": 2.5622977171066155,  
    "K_us": 0.002745633390344498,  
    "b": 0.00039676132180812505,  
    "tau": -0.02283428861943423  
  },  
  "HYUNDAI_IONIQ_5": {  
    "L_eff": 3.0387877407606987,  
    "K_us": 0.004732031059157526,  
    "b": 0.0019402727789656954,  
    "tau": -0.038122141286844304  
  },  
  "TESLA_MODEL_3": null  
}
```

module-2.v3_agent-04

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "L_eff": 3.96394551474604,
    "Kus": 0.003414866921617812,
    "bias": -0.005292773687032753,
    "tau": -0.06157679065853846
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "L_eff": 2.569483805164948,
    "Kus": 0.0025788290081397765,
    "bias": 0.0011230859100703557,
    "tau": -0.04709908110775255
  },
  "HYUNDAI_IONIQ_5": {
    "L_eff": 3.008377713685593,
    "Kus": 0.005133658670161866,
    "bias": 0.0019487320168033527,
    "tau": -0.04867295794401198
  },
  "TESLA_MODEL_3": {
    "dummy": 0.0
  }
}
```

module-2.v3_agent-06

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "platform": "FORD_F_150_LIGHTNING_MK1",
    "L": 3.818759956786134,
    "Kus": 0.0039432563422251285,
    "tau": -0.05861405092290705,
    "bias": -0.004437268567086911,
    "rmse_fit": 0.0054979371675811595,
    "n": 430990,
    "converged": true
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "platform": "FORD_MUSTANG_MACH_E_MK1",
    "L": 2.4997194805464074,
    "Kus": 0.0024024802542751437,
    "tau": -0.06630813663570458,
    "bias": 0.00022227158439854945,
    "rmse_fit": 0.00894166094428278,
    "n": 615175,
    "converged": true
  },
  "HYUNDAI_IONIQ_5": {
    "platform": "HYUNDAI_IONIQ_5",
    "L": 3.100074393615448,
    "Kus": 0.0035463766130646316,
    "tau": -0.05358525493540066,
    "bias": 0.001966636983251082,
    "rmse_fit": 0.008266374104865987,
    "n": 2042270,
    "converged": true
  }
}
```


module-2.v3_agent-08

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "K_us": 0.004622784347956018,
    "tau": -0.05460138412592134,
    "bias": -0.004482561615678892,
    "train_mse": 3.3520085701320475e-05,
    "train_rmse": 0.005789653331704799,
    "n_samples": 430990,
    "n_segments": 164,
    "converged": true,
    "message": "CONVERGENCE: NORM OF PROJECTED GRADIENT <= PGTOL"
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.984,
    "K_us": 0.0008351228965168481,
    "tau": -0.09459794094766563,
    "bias": 0.00054055263346224,
    "train_mse": 0.00017605613840148047,
    "train_rmse": 0.01326861478834473,
    "n_samples": 615175,
    "n_segments": 232,
    "converged": true,
    "message": "CONVERGENCE: NORM OF PROJECTED GRADIENT <= PGTOL"
  },
  "HYUNDAI_IONIQ_5": {
    "L": 2.95,
    "K_us": 0.004646731366208564,
    "tau": -0.04946605447751251,
    "bias": 0.0018472769568295111,
    "train_mse": 7.578893405982132e-05,
    "train_rmse": 0.008705684008727937,
    "n_samples": 2042270,
    "n_segments": 785,
    "converged": true,
  }
}
```

```
    "message": "CONVERGENCE: NORM OF PROJECTED GRADIENT <= PGTOL"
  }
}
```

module-2.v3_agent-10

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "s_d": 0.9300245172021879,
    "c_d": 0.347294150023435,
    "tau_d": -0.05773764830098028,
    "K_us": 0.003240471803333766,
    "b": -0.004299834214218103,
    "L": 3.7
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "s_d": 1.1508056993666387,
    "c_d": 0.7498585082419895,
    "tau_d": -0.07607250144234215,
    "K_us": 0.0022785419555576775,
    "b": 0.0003167396785048121,
    "L": 2.984
  },
  "HYUNDAI_IONIQ_5": {
    "s_d": 0.9422644146648296,
    "c_d": 0.3465308670029774,
    "tau_d": -0.050503079814885346,
    "K_us": 0.002862544790766583,
    "b": 0.0022556662651586025,
    "L": 3.0
  },
  "TESLA_MODEL_3": {
    "passthrough": true
  }
}
```

module-3.v2_agent-02

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 0.8599716540800916,
    "L_eff": 3.260802128118046,
    "K_us": 0.0033605485538025595,
    "tau": 0.058798336746219135,
    "delta0": 0.0012445223286377857,
    "use_per_segment_delta0": false
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 0.8948751688442338,
    "L_eff": 2.218474888909999,
    "K_us": 0.002002066992546276,
    "tau": 0.06321825538415474,
    "delta0": -0.0001,
    "use_per_segment_delta0": true
  },
  "HYUNDAI_IONIQ_5": {
    "g": 0.928157495232038,
    "L_eff": 2.890194195032293,
    "K_us": 0.002716802199963435,
    "tau": 0.051288867659338114,
    "delta0": 0.0005262026839657872,
    "use_per_segment_delta0": true
  }
}
```

module-3.v2_agent-04

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 0.8627559173927216,
    "L_eff": 3.2661642837279934,
    "K_us": 0.003399629365556295,
    "tau": 0.05764217451751205,
    "delta0": 0.0011797692301086372,
    "delta0_fallback": 0.0011797692301086372,
    "use_per_segment_delta0": false,
    "train_rmse": 0.005285188367801501,
    "dev_rmse": 0.006580546885659461,
    "bias_spread_std": null
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 1.2845766366025777,
    "L_eff": 3.185078856835661,
    "K_us": 0.002783912493950955,
    "tau": 0.06245560987504152,
    "delta0": -0.0019790637549735985,
    "delta0_fallback": -0.0019790637549735985,
    "use_per_segment_delta0": true,
    "train_rmse": 0.008444302053948288,
    "dev_rmse": 0.008286227301504138,
    "bias_spread_std": null
  },
  "HYUNDAI_IONIQ_5": {
    "g": 0.9454031576502736,
    "L_eff": 2.9345636591691626,
    "K_us": 0.0028198034203513055,
    "tau": 0.0507163687871907,
    "delta0": 0.0004041780050478772,
    "delta0_fallback": 0.0004041780050478772,
    "use_per_segment_delta0": true,
    "train_rmse": 0.007930751971921808,
    "dev_rmse": 0.00660747263955304,
    "bias_spread_std": null
  }
}
```

module-3.v2_agent-06

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "use_per_segment_delta0": false,
    "delta0": 0.00133,
    "g": 0.863,
    "L_eff": 3.26,
    "K_us": 0.0035,
    "tau": 0.06
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "use_per_segment_delta0": true,
    "delta0_fallback": -0.0001,
    "g": 0.891,
    "L_eff": 2.22,
    "K_us": 0.0015,
    "tau": 0.069
  },
  "HYUNDAI_IONIQ_5": {
    "use_per_segment_delta0": true,
    "delta0_fallback": 0.0,
    "g": 0.938,
    "L_eff": 2.887,
    "K_us": 0.00289,
    "tau": 0.062
  }
}
```

module-3.v2_agent-08

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "use_per_segment_delta0": false,
    "delta0": 0.00133,
    "g": 0.863,
    "L_eff": 3.26,
    "K_us": 0.0035,
    "tau": 0.06
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "use_per_segment_delta0": true,
    "delta0_fallback": -0.0001,
    "g": 0.891,
    "L_eff": 2.22,
    "K_us": 0.0015,
    "tau": 0.069
  },
  "HYUNDAI_IONIQ_5": {
    "use_per_segment_delta0": true,
    "delta0_fallback": 0.0,
    "g": 0.938,
    "L_eff": 2.887,
    "K_us": 0.00289,
    "tau": 0.062
  }
}
```

module-3.v2_agent-10

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "use_per_segment_delta0": false,
    "delta0": 0.00133,
    "g": 0.863,
    "L_eff": 3.26,
    "K_us": 0.0035,
    "tau": 0.06
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "use_per_segment_delta0": true,
    "delta0_fallback": -0.0001,
    "g": 0.891,
    "L_eff": 2.22,
    "K_us": 0.0015,
    "tau": 0.069
  },
  "HYUNDAI_IONIQ_5": {
    "use_per_segment_delta0": true,
    "delta0_fallback": 0.0,
    "g": 0.938,
    "L_eff": 2.887,
    "K_us": 0.00289,
    "tau": 0.062
  }
}
```

module-3.v3_agent-02

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "a": 0.9839278434685126,
    "b": -0.0004596324959947,
    "c": 0.0003973122452781526,
    "d": -0.007997476152353435
  },
}
```

```
"FORD_MUSTANG_MACH_E_MK1": {  
  "a": 0.9727773793773095,  
  "b": 0.0016981202953789607,  
  "c": 0.0001999394762639504,  
  "d": -0.02207103010528879  
},  
"HYUNDAI_IONIQ_5": {  
  "a": 0.9913665028013989,  
  "b": 0.0006388079681906442,  
  "c": 0.00014836133647526484,  
  "d": 0.00450672562112523  
}  
}
```


module-3.v3_agent-06

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "w": [
      0.039210004724697683,
      0.0027898060882411775,
      -8.213353932869296e-05,
      -0.002101385572330489,
      0.015375393428030156,
      -0.019248388299671407,
      0.0017437669476709928
    ],
    "features": [
      "yr_v1",
      "abs_yr_v1",
      "v",
      "v*yr_v1",
      "ddelta_dt",
      "delta",
      "1"
    ],
    "lambda": 30.0
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "w": [
      0.04786803033956984,
      0.0010252169296772547,
      -0.0001550342759403376,
      0.0007801849572229949,
      0.03973697702371263,
      -0.04132919662285873,
```

```
0.0029539513418850413
],
"features": [
  "yr_v1",
  "abs_yr_v1",
  "v",
  "v*yr_v1",
  "ddelta_dt",
  "delta",
  "1"
],
"lambda": 30.0
},
"HYUNDAI_IONIQ_5": {
  "w": [
    0.11163643144945354,
    -0.01144073312700024,
    -8.938098452332296e-05,
    -0.0043289766474131726,
    0.011781066593348639,
    -0.1397503946290344,
    0.0011078590121685729
  ],
  "features": [
    "yr_v1",
    "abs_yr_v1",
    "v",
    "v*yr_v1",
    "ddelta_dt",
    "delta",
    "1"
  ],
  "lambda": 30.0
}
```

module-3.v3_agent-08

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "features": [
      "bias",
      "delta_road",
      "v",
      "v_delta",
      "ddelta",
      "v_ddelta",
      "v0_yaw",
      "abs_v0_yaw",
      "a_long",
      "v_sq_delta"
    ],
    "beta": [
      -0.0013823839722696969,
      0.04137416440068437,
      7.448672929471383e-05,
      -0.012310248841363941,
      -0.006256706876683462,
      -0.0013792677587728672,
      0.0045077768608475025,
      -0.004229129267950782,
      0.00046053718010623685,
      0.00047676588153397785
    ],
    "n_samples": 430990
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "features": [
      "bias",
      "delta_road",
      "v",
      "v_delta",
      "ddelta",
      "v_ddelta",
      "v0_yaw",
      "abs_v0_yaw",
      "a_long",
      "v_sq_delta"
    ],
    "n_samples": 430990
  }
}
```

```

    "beta": [
      -0.0002765413679144323,
      0.04444765601029079,
      7.361683614844277e-05,
      -0.010525182430347446,
      0.0006366996373336289,
      -0.003027149663796255,
      0.004991171014731363,
      0.01051162443915557,
      -0.0003707902497743102,
      -0.0004177085900277482
    ],
    "n_samples": 615175
  },
  "HYUNDAI_IONIQ_5": {
    "features": [
      "bias",
      "delta_road"
    ]
  }
}

```

module-3.v3_agent-10

```

{
  "FORD_F_150_LIGHTNING_MK1": {
    "intercept": -0.0015292016694573706,
    "weights": {
      "v_mps": 7.523513452594086e-05,
      "delta_road_rad": 0.015247619064453013,
      "d_delta_dt": -0.008088882179724404,
      "a_long_mps2": 0.0004266683248191665,
      "yr_v1": -0.010938174180205732,
      "abs_delta": -0.0046619127252169905,
      "yr_v1_sq_signed": 0.0031131630350753935
    }
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "intercept": -0.00025671546619143265,
    "weights": {
      "v_mps": 7.597710383706725e-05,

```

```

    "delta_road_rad": 0.02230237156926512,
    "d_delta_dt": -0.004979903507745553,
    "a_long_mps2": -0.0003707521334163227,
    "yr_v1": -0.025822216062859112,
    "abs_delta": 0.022243029033033466,
    "yr_v1_sq_signed": 0.012709323274160847
  }
},
"HYUNDAI_IONIQ_5": {
  "intercept": -7.400008140931345e-06,
  "weights": {
    "v_mps": 3.4765997046644015e-05,
    "delta_road_rad": 0.007171887290213692,
    "d_delta_dt": 9.749220428357943e-05,
    "a_long_mps2": 0.0006747858420646872,
    "yr_v1": -0.010175331098695933,
    "abs_delta": 0.009304240486557539,
    "yr_v1_sq_signed": 0.00642116668196153
  }
}
}

```

module-4.v1.01_agent-03

```

{
  "TESLA_MODEL_3": {
    "bias": 0.0,
    "ridge_w": [
      0.0,
      0.0,

```

```
    0.0,
    0.0,
    0.0,
    0.0
],
"lambda": 1000.0,
"feat_mu": [
    0.0013746071886924117,
    0.028929884623248848,
    19.099479879654037,
    -0.003983509528507812,
    0.012989695599658083,
    0.00022546668940797497
],
"feat_sigma": [
    0.04556600472011049,
    0.21361830215681743,
    10.91183950057269,
    0.6714085412983767,
    0.04369688931226369,
    0.04314597543455112
]
},
"FORD_F_150_LIGHTNING_MK1": {
    "bias": 0.0,
    "ridge_w": [
        0.0,
        0.0,
        0.0,
        0.0,
        0.0,
        0.0
    ],
    "lambda": 1000.0,
    "feat_mu": [
        -0.0015137800038034729,
        -0.0037159634576037787,
        16.300011855626813,
        0.12540514439432449,
        0.01810040392970318,
        0.0008774824950494891
    ],
    "feat_sigma": [
```

```
0.0636911953265782,  
0.28931656976699976,  
10.717432158380031,  
12.537011944228352,  
0.061083837996636484,  
0.08104849692314077  
],  
"_disabled_reason": "cohort \u00a75: Lightning yaw saturates &
```

module-4.v1_agent-01

```
{  
  "ridge_lambda": 30.0,  
  "feature_names": [  
    "const",  
    "delta_road",  
    "abs_delta_road",  
    "v",  
    "delta*v",  
    "abs_delta*v",  
    "delta^2*v",  
    "d_delta_dt",  
    "yr_v1"  
  ],  
  "platforms": {  
    "FORD_MUSTANG_MACH_E_MK1": {  
      "bias_only": 0.0013360177352628089,  
      "beta": [  
        0.00018685026009305166,  
        0.09219828994588719,  
        -0.025783629417111403,  
        7.5439538414678e-05,  
        0.0017876040006721155,  
        -0.007296337051731339,  
        0.09478364424398987,  
        -0.021146729244488356,  
        -0.06576959206990736  
      ],  
    },  
  },  
}
```

```
"cv_residual_rmse": 0.011656881352946322,  
"cv_residual_rmse_std": 0.007734736333138539,  
"cv_baseline_rmse": 0.011999142746660713  
},  
"HYUNDAI_IONIQ_5": {  
  "bias_only": 0.0006812541510197226,  
  "beta": [  
    -0.0006780931189280977,  
    0.06344781976387885,  
    0.01591619487417097,  
    7.160187898114387e-05,  
    0.038590284408058076,  
    0.0024337623533112877,  
    -0.009040102750581378,  
    -0.016988454521445543,  
    -0.16791095173730144  
  ],  
  "cv_residual_rmse": 0.007091703321227194,  
  "cv_residual_rmse_std": 0.00185153915165244
```

module-4.v1_agent-04

```
{  
  "FORD_F_150_LIGHTNING_MK1": {  
    "apply": true,  
    "intercept": -0.0018706929091562578,  
    "features": [  
      "delta",  
      "v_delta",  
      "v2_delta",  
      "yr_v1",  
      "v_yr_v1",  
      "a_long"  
    ],  
    "coefs": [  
      -0.10259858490293011,  
      0.03155541535607884,  
      0.0016150237555660578,
```



```

    0.004907942076633252,
    -0.016907765297495773,
    -1.180589496058453e-06
  ],
  "cv_v1_rmse": 0.011421948177936606,
  "cv_head_rmse": 0.011239609732498611,
  "feature_set": "rich",
  "lambda": 100.0
},
"FORD_MUSTANG_MACH_E_MK1": {
  "apply": false,
  "intercept": 0.0013128122793277302,
  "features": [
    "delta",
    "v_delta",
    "v2_delta",
    "yr_v1",
    "v_yr_v1",
    "a_long"
  ],
  "coefs": [
    -0.13405601646838025,
    0.03757262372233324,
    0.001144751740135907,
    0.001509552370891812,
    -0.011712471653399673,
    -0.0003404605240783607
  ],
  "cv_v1_rmse": 0.0119345532668796,
  "cv_head_rmse": 0.011955169146202352,
  "feature_set": "rich",
  "lambda": 100.0
},
"HYUNDAI_IONIQ_5": {
  "apply": true,
  "intercept": 0.0009697876465239235,
  "features": [
    "delta",
    "

```

module-4.v1_agent-07

```
{
  "bias": {
    "TESLA_MODEL_3": 2.999132956583475e-10,
    "FORD_F_150_LIGHTNING_MK1": -0.0041132124076757445,
    "FORD_MUSTANG_MACH_E_MK1": -4.1278547737791775e-05,
    "HYUNDAI_IONIQ_5": 0.003523757621288774
  },
  "ridge": {
    "TESLA_MODEL_3": {
      "lambda": 10000.0,
      "feature_names": [
        "delta",
        "abs_delta",
        "v",
        "delta_v",
        "delta_abs_delta",
        "a_long",
        "delta_sq",
        "sign_delta_v"
      ],
      "weights_raw": [
        -5.9197077715805255e-08,
        -1.0015789694889157e-07,
        5.530281265597394e-11,
        1.5142645564086152e-08,
        1.208444849458761e-08,
        1.6620711765189437e-10,
        1.700549887724333e-07,
        -5.200159654692222e-11
      ],
      "intercept_raw": -2.773044472289003e-10,
      "holdout_mse": 2.6719543584301285e-14,
      "n_train_samples": 527433
    },
    "FORD_F_150_LIGHTNING_MK1": {
      "lambda": 10.0,
      "feature_names": [
        "delta",
        "abs_delta",
        "v",
        "delta_v",
```

```
    "delta_abs_delta",
    "a_long",
    "delta_sq",
    "sign_delta_v"
  ],
  "weights_raw": [
    0.3410909623623404,
    0.00
```

module-4.v2_agent-08

```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 0.8808038763255166,
    "L_eff": 2.2237793178633387,
    "K_us": 0.0017632231957256993,
    "tau": 0.02732363043764696,
    "delta0_fallback": -0.005312458941976969,
    "alpha_steer_rate": -0.10183367313108851,
    "use_per_segment_delta0": true
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 0.8469022562788538,
    "L_eff": 3.2643013153219167,
    "K_us": 0.003145185602493612,
    "tau": 0.03507802069651487,
    "delta0_fallback": 0.0012105417102480814,
    "alpha_steer_rate": -0.03610959850891402,
    "use_per_segment_delta0": false
  },
  "HYUNDAI_IONIQ_5": {
    "g": 0.9269585851786253,
    "L_eff": 2.894242280397565,
    "K_us": 0.0026380815832284773,
    "tau": 0.05309874948850927,
    "delta0_fallback": 0.0016114738881128556,
    "alpha_steer_rate": 0.0024241400357318883,
```

```
    "use_per_segment_delta0": true
  }
}
```

module-4.v2_agent-10

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 0.8225565786260306,
    "L_eff": 3.270761500568214,
    "K_us": 0.0027280391940200334,
    "tau": 0.023124512232105744,
    "K_us2": -0.01553611821787546,
    "bias": 0.00029081942067580853,
    "k_ff": -0.008879084366667647,
    "use_per_segment_delta0": false,
    "delta0": 0.00133,
    "delta0_fallback": 0.0
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 0.8099226898614635,
    "L_eff": 2.2545626972209125,
    "K_us": 0.0005836732051187025,
    "tau": 0.04490805042479574,
    "K_us2": -0.18601413847441528,
    "bias": 0.002801296861051317,
    "k_ff": -0.007941883417939366,
    "use_per_segment_delta0": true,
    "delta0": 0.0,
    "delta0_fallback": -0.0001
  },
  "HYUNDAI_IONIQ_5": {
    "g": 0.8656852935099641,
    "L_eff": 2.910675491855111,
    "K_us": 0.001193590623662968,
    "tau": 0.01743602227795721,
    "K_us2": -0.08204270923185958,
```

```
"bias": 0.001220432190762801,  
"k_ff": -0.009103659740814939,  
"use_per_segment_delta0": true,  
"delta0": 0.0,  
"delta0_fallback": 0.0  
}  
}
```

- **6b. Operating-contract audit**

The grader strips `sim_df` to an allowlist before calling each agent's `predict()`, so an agent cannot read truth at inference time. This section reports source-level intent — which agents *would have* read truth columns if the grader hadn't intervened. A clean audit is its own quality signal.

Clean cohort: no agent referenced truth-derived column names inside their `predict()` body.

7. Reconstruction quality (substrate signal)

How many agents shipped the right artefacts to be canonically gradable. Failures here are a substrate or contract problem, not a model problem.

format check	pass	fail
agent_folder_exists	105	0
has_manifest_json	94	11
manifest_parsable	94	11
manifest_declares_predict_callable	94	11
manifest_declares_platform_support	94	11
has_predict_py	94	11
has_coeffs_json	73	32
has_report	33	72

Failure reasons across the cohort:

- missing_manifest_json — 11 agent(s)

- `import_failed` — 1 agent(s)

• 8. Worst-of-cohort

Lowest yaw $\Delta\%$

- `module-4.v1_agent-10`
(+39.3%)
- `m1-agent-08`
(+40.4%)
- `module-4.v1_agent-07`
(+44.6%)

Lowest CTE $\Delta\%$

- `module-4.v1_agent-10`
(+36.6%)
- `m1-agent-10`
(+49.5%)
- `m2-agent-07`
(+49.5%)